

Name: _____

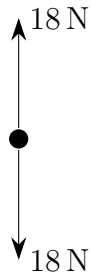
Date: _____

Period: _____

Newton #1 (Net Force)

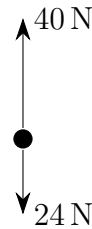
1. In each of the free-body diagrams below, calculate the **magnitude** and **direction** of the net force and draw it.

(a)



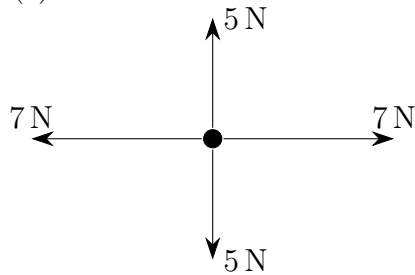
$$F_{NET} = \text{_____ N, _____}$$

(b)



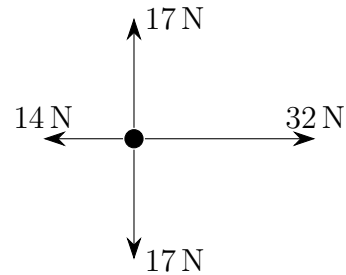
$$F_{NET} = \text{_____ N, _____}$$

(c)



$$F_{NET} = \text{_____ N, _____}$$

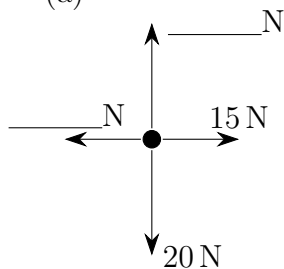
(d)



$$F_{NET} = \text{_____ N, _____}$$

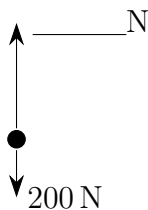
2. In each of the free-body diagrams below, the net force is given, but one or more of the applied forces is missing. Find the missing forces.

(a)



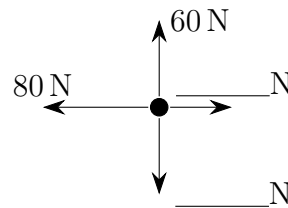
$$F_{NET} = 0 \text{ N}$$

(b)



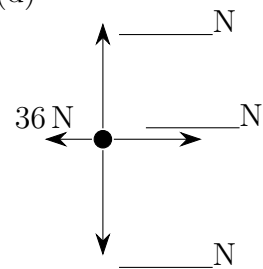
$$F_{NET} = 150 \text{ N, up}$$

(c)



$$F_{NET} = 45 \text{ N, left}$$

(d)



$$F_{NET} = 23 \text{ N, right}$$

Name: _____

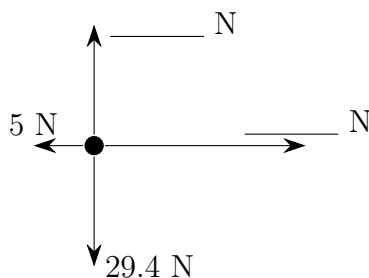
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3. What is the acceleration of a 1500-kg car that experiences a net force of 970 N?

4. Fill in the blanks in each of the situations depicted below. Draw the net force.

(a)

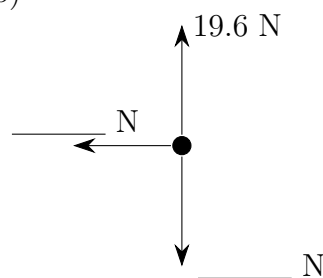


$$m = 3 \text{ kg}$$

$$a = \text{ } \text{m/s}^2, \text{ } \text{ }$$

$$F_{NET} = 23 \text{ N, right}$$

(b)

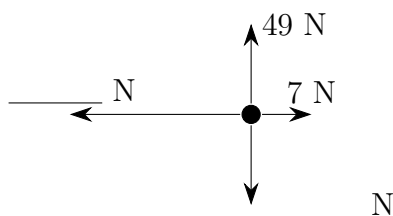


$$m = 2 \text{ kg}$$

$$a = 8 \text{ m/s}^2, \text{ left}$$

$$F_{NET} = \text{ } \text{N, } \text{ }$$

(c)

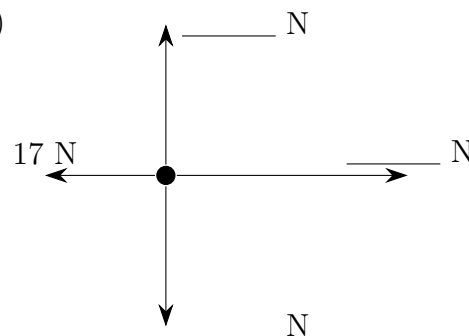


$$m = 5 \text{ kg}$$

$$a = 12 \text{ m/s}^2, \text{ left}$$

$$F_{NET} = \text{ } \text{N, } \text{ }$$

(d)



$$m = 3 \text{ kg}$$

$$a = 18 \text{ m/s}^2, \text{ right}$$

$$F_{NET} = \text{ } \text{N, } \text{ }$$

5. An airplane has a mass of 2500 kg. It needs to get up to a speed of 30 m/s in order to take off. How much net force is needed to get the plane from rest up to this speed on a 245 m runway? (Hint: *Begin by finding acceleration!*)